Advanced Algebra & Systems

Section 1: Polynomial Division & Remainder Theorem

1 When the polynomial $P(x) = 2x^3 - 5x^2 + kx - 8$ is divided by $(x - 3)$, the remainder is 10. What is the value of k?

a. 3

b. 5

c. 7

d. 9

2 What is the quotient when $4x^3 + 2x^2 - 6x + 5$ is divided by $(2x + 1)$?

a. $2x^2 - 3$

b. $2x^2 + x - 3$

c. $2x^2 - 3x + 1$

d. $2x^2 + 3$

3 If $(x + 2)$ and $(x - 1)$ are both factors of $x^3 + ax^2 + bx - 6$, what is the value of $a + b$?

a. -1

b. 1

c. 3

d. 5

Section 2: Vieta's Formulas & Symmetric Sums

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4 Let r and s be the roots of $2x^2 - 6x + 1 = 0$. What is the value of $r^2 + s^2$?

a. 7

b. 8

c. 9

d. 10

5 Let p, q, r be the roots of $x^3 - 4x^2 + 2x - 7 = 0$. What is the value of $\frac{1}{p} + \frac{1}{q} + \frac{1}{r}$?

a. $\frac{2}{7}$

b. $-\frac{2}{7}$

c. $\frac{4}{7}$

d. $-\frac{4}{7}$

6 If the roots of $x^2 - kx + 36 = 0$ are in the ratio $1 : 4$, what is the positive value of k?

a. 12

b. 15

c. 18

d. 20

Section 3: Non-Linear Systems of Equations

7 Find the positive y-value of the points of intersection of $x^2 + y^2 = 25$ and $y = x^2 - 5$.

a. 0

b. 3

c. 4

d. 5

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8 How many real solutions (x, y) satisfy the system $xy = 6$ and $x^2 + y^2 = 13$?

a. 1

b. 2

c. 4

d. 0

9 In the system $y = 2x + k$ and $y = -x^2 + 4x - 1$, for what value of k are the lines tangent?

a. 0

b. 1

c. 2

d. 3

Section 4: Systems of Three Variables & Matrices

10 Solve the system for z: $x + y + z = 6$, $2x - y + z = 3$, and $x + 2y - z = 2$.

a. 1

b. 2

c. 3

d. 4

11 For what value of a will the system $x + y = 3$, $2x + 2y = a$ have no solution?

a. Any $a \neq 6$

b. $a = 6$

c. $a = 0$

d. $a = 3$

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12 What is the value of x in the unique solution to the matrix equation $[[2, 1], [3, 4]][[x], [y]] = [[7], [18]]$?

a. 1

b. 2

c. 3

d. 4

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